



Cloud & IoT Monitoring and Data analysis with Elasticsearch and Grafana

Technical Reflection

Bachelor's degree in the Applied Computer
Science

Birate Kabanza Arthur

Academic year 2024-2025

Campus Geel, Kleinhoefstraat 4, BE-2440 Geel

1. Introduction

The technical reflection demonstrates my work accomplishments, the knowledge I gained, and the technical challenges I faced during my Van Genechten Packaging internship. The document defines the project scope and the technical progress I achieved while highlighting the business value of my contributions. It illustrates how I applied theoretical concepts in real-world situations and details my problem-solving techniques for obstacles, as well as the practical knowledge I acquired for future technical tasks.

2. Project Achievements

The internship allowed me to contribute to multiple areas of infrastructure monitoring, analytics, and system performance optimization. Key achievements include:

Dashboard Development in Grafana and Kibana

Designed multiple real-time dashboards that visualized infrastructure health, backup performance, and IoT device activity.

Configured panels that tracked CPU usage, memory consumption, error logs, and application response times. Introduced role-specific views: system administrators could monitor servers, while developers had access to logs and performance metrics relevant to applications.

Data Pipeline Setup with Logstash

Established robust data pipelines to ingest log and performance data from multiple sources into Elasticsearch. Configured parsing rules to handle inconsistent and unstructured log data, transforming it into structured, queryable fields.

Optimized pipeline performance by filtering unnecessary fields, which reduced storage requirements and improved query speed.

Anomaly Detection with Elasticsearch Machine Learning

Configured an unsupervised anomaly detection system to identify unusual patterns in log data, such as unexpected spikes in error rates or system failures.

Supported predictive maintenance by allowing early detection of potential breakdowns in infrastructure and IoT devices.

Comprehensive Monitoring Coverage

Covered all key infrastructure layers:

- Virtual Machines: Monitored uptime, resource consumption, and health.
- Applications: Captured logs, errors, and performance bottlenecks.
- Backup Systems: Ensured that data backups were running on schedule and without failure.
- IoT Devices: Integrated IoT monitoring for devices used in packaging operations.

This comprehensive monitoring provided Van Genechten Packaging with a single view of both on-premises and cloud-based systems.

2. Value for the Company

The project provided clear technical and business value for Van Genechten Packaging:

Early Detection of Anomalies:

The dashboards and anomaly detection features enabled faster detection of irregularities, reducing downtime and increasing overall reliability.

Real-Time Visibility:

Provided continuous insights into the state of systems and infrastructure, which previously required manual checks or fragmented tools.

Operational Efficiency:

Monitoring systems automate many tasks that previously required manual intervention, saving time and reducing the chance of human error.

Foundation for Future Growth:

The tools and processes put in place laid the foundation for expanding monitoring practices to more plants, applications, and business units.

3. Completion Status and Future Work

Completed Deliverables:

- Dashboards for virtual machines, applications, IoT activity, and backup performance.
- Logstash pipelines to ingest and standardize log data.
- Basic anomaly detection system deployed in Elasticsearch.

Pending Work:

- The iDRAC monitoring dashboard (for server health monitoring) could not be completed due to data ingestion issues. Several critical fields were missing or incomplete, which limited visibility.

Future Recommendations:

- Centralized Documentation: Before future projects, create clear documentation of expected dashboards, KPIs, and data sources. This ensures that stakeholders' needs are aligned with the technical implementation.
- Standardized Data Pipelines: Enforce consistency in data sources to minimize integration challenges.
- Scalability Planning: As the monitoring environment grows, additional optimizations (e.g., data archiving policies, distributed queries) will be required.
- Automation of Alerts: More advanced alerting and automated responses should be integrated to reduce reliance on manual monitoring.

4. Challenges Faced

The internship also highlighted common challenges in building and scaling monitoring solutions:

Unclear Requirements: Some requested KPIs were vaguely defined, which required additional clarification. This prompted me to enhance my communication with stakeholders and improve documentation.

Incomplete Data Sources: Several dashboards could not be fully developed due to missing fields. This emphasized the importance of validating data availability before development.

Performance Bottlenecks: Handling large volumes of log data occasionally caused Elasticsearch queries to slow down. Research and experimentation with indexing strategies helped mitigate the issue.

Balancing Simplicity and Detail: Dashboards risked becoming too cluttered. I learned to focus on actionable insights rather than trying to visualize every metric.

.

6. Conclusion

This internship was a valuable technical experience that allowed me to work on real-time monitoring and analytics at scale. By developing dashboards, optimizing data pipelines, and applying anomaly detection, I contributed to a stronger monitoring environment for Van Genechten Packaging.

The project not only provided practical technical skills but also improved my ability to handle unclear requirements, deal with imperfect data, and design solutions that provide tangible business value. These experiences will be highly relevant in my future career in data engineering, analytics, or cloud infrastructure monitoring.